Characterization of the Short-Range Couplings in Excitation Energy Transfer

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The electronic coupling in excitation energy transfer (EET) is composed of a Coulomb coupling (which can be approximated as the Förster's dipole coupling), a Dexter's exchange coupling, and a term arising from the orbital overlap of the donor and acceptor fragments. We have developed a new scheme to account for the EET coupling by generalizing the fragment charge difference scheme [Voityuk, A. A.; Rösch, N. *J. Chem. Phys.* **2002**, *117*, 5607] for electron-transfer coupling. As a result, the EET coupling can be calculated in a general class of systems irrespective of the molecular symmetry. The short-range coupling, defined as the contribution from Dexter's exchange coupling and the overlap effect, was obtained as the difference of the EET coupling from this new scheme and a precise account of the Coulomb coupling. For a pair of stacked naphthalenes, the short-range coupling is very similar to the triplet−triplet energy-transfer coupling in both magnitudes and the distance dependences. We have observed cases in which the Coulomb coupling decays either faster or slower than the expected dipole−dipole R^{-3} distance dependence even at distances of 10−20 Å, due higher multipole interaction. For a well-studied series of rigidly linked naphthalene dimers, the role of through-bond interaction was studied with the new computational methods. Similar to previous reports, our results show that the through-bond component contains a Coulomb coupling, and in some cases, contributions from the short-range couplings can be seen.

1. Introduction

Electronically excited molecules may pass their excitation energy to another molecule through the excitation energy transfer (EET).[1] EET is seen in both artificial[2−6] and natural light-harvesting systems,[7] as well as the fluorescence resonance energy transfer (FRET), a technique which has been widely used in many different areas including bioanalysis,[8] single-molecule fluorescence studies,[9,10] molecular photonic wires,[11,12] and molecular optoelectronic gates.[13]

In EET, it can be shown that the electronic coupling contains three terms under a first-order approximation, the Coulomb coupling, which is the Coulomb interaction between electronic transitions, Dexter exchange coupling that accounts for the indistinguishability of the electrons in many-electron wave functions, and a term arising from the overlap of donor−acceptor electronic densities.[14−19] Coulomb coupling has been regarded as the major contribution of EET, but recently, there exists an intense interest in experimentally characterizing the short-range components of EET coupling, including the exchange and overlap effects.[20,21] It is fundamentally important to theoretically characterize the size of the exchange coupling and to see if a system with a strong exchange or short-range coupling can be found because the short-range couplings may provide a handle to design molecules with desired properties.

The distance dependence of several bridge-mediated EETs has been reported, with a deviation from a typical dipole−dipole form in short distances.[20−22] At short distances, both the Dexter and overlap terms are expected. However, the dipole approximation may also break down in short donor−acceptor distances, and the full Coulomb coupling may not follow a typical R^{-3} dipole−dipole form. The spherical harmonic approximation may not be valid, or the effects of higher-order multipole terms may be important.[23,24] Therefore, observation of a distance dependence that is steeper than R^{-3} is not sufficient to conclude the role of the short-range couplings.

To help resolve the difficulty of deducing the origin of electronic coupling from experimental data, it is important to find reliable theoretical estimates of the magnitude of the short-range coupling strength. Theoretical accounts of the EET coupling have been mostly limited to the calculation of either the dipole−dipole coupling,[25] multipole coupling,[26] or the full Coulomb interaction.[23,24] In these calculations, the contributions of the exchange and overlap terms are ignored. The Coulomb and the exchange-correlation interactions were calculated for symmetric systems in a recent work.[27] Up until now, there has not been yet a report on the strength of the short-range coupling. In the present work, we characterize both the total EET coupling and the Coulomb coupling in a good numerical quality and use their difference as an estimate for the short-range EET components. To the best of our knowledge, it is the first time such short-range couplings for EET were analyzed, and our results could shed light on the development of controlling the EET processes through molecular design.

Under the avoided-crossing condition, where the two "uncoupled" zero-order states are at the same energy, the full EET coupling can be calculated as half of the energy gap of a pair of excitonic states.[27−29] For a symmetric system with identical donor and acceptor fragments, the avoided-crossing condition can be easily satisfied. However, for more commonly seen

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asymmetric systems, it is hard to meet such a condition.[30,31] Therefore, it is desirable to develop new schemes that can account for the full EET coupling for a general class of systems.

On the other hand, in calculating electron-transfer (ET) coupling, the generalized Mulliken−Hush (GMH)[32] method uses eigenstate properties, and it is suitable for a general class of systems. The charge-localized states are linear combinations of these two eigenstates, under the condition of a zero transition dipole between the two charge-localized states. The ET coupling is expressed in terms of the state energy difference, dipole moment difference, and the transition dipole between the two states, all calculated in the eigenstate space. On the basis of a similar strategy, the fragment charge difference (FCD) scheme calculates the off-diagonal Hamiltonian matrix element in the space that has an optimal charge separation.[33]

In this paper, we report the development of a new method to calculate EET coupling. It is an extension of the FCD scheme. We have defined an excitation density operator as the sum of the electron and hole densities created in the excitation. The excitation density is used in place of the charge density of FCD. Since eigenstate properties are used, the EET coupling obtained reflects the full coupling under the level of the theory employed for the eigenstates. With a precise account of the Coulomb coupling, we have characterized the short-range coupling, which includes both the exchange and the overlap contributions. Our results indicate that the short-range terms in EET behave very similarly to triplet−triplet energy-transfer coupling,[34] a spin-exchange energy transfer that does not contain Coulomb coupling. For a series of well-studied rigidly linked naphthalene molecules, we demonstrate that the nature of through-bond coupling contains a Coulomb element, while the short-range coupling contributed significantly in some of the cases studied.

2. Theory and Methods

Under a first-order approximation in the single excitation theory, the excitation energy-transfer coupling of two molecular fragments was shown to be[17,18]

$$J^0 = J_{\mathrm{coul}} + J_{\mathrm{ex}} + J_{\mathrm{overlap}} \tag{1}$$

where J_{coul}, J_{ex}, and J_{overlap} are defined as

$$J_{\mathrm{coul}} = \int \mathrm{d}\mathbf{r} \int \mathrm{d}\mathbf{r}' \rho_{\mathrm{D}}^{*}(\mathbf{r}) \frac{1}{|\mathbf{r} - \mathbf{r}'|} \rho_{\mathrm{A}}(\mathbf{r}') \tag{2}$$

$$J_{\mathrm{ex}} = \int \mathrm{d}\mathbf{r} \int \mathrm{d}\mathbf{r}' \rho_{\mathrm{D}}^{*}(\mathbf{r},\mathbf{r}') \frac{1}{|\mathbf{r} - \mathbf{r}'|} \rho_{\mathrm{A}}(\mathbf{r}',\mathbf{r}) \tag{3}$$

$$J_{\mathrm{overlap}} = -\omega_0 \int \mathrm{d}\mathbf{r} \rho_{\mathrm{D}}^{*}(\mathbf{r}) \rho_{\mathrm{A}}(\mathbf{r}) \tag{4}$$

in which the donor and acceptor excitation may not be orthogonal. The $\rho_{\mathrm{D}}(\mathbf{r})$ and $\rho_{\mathrm{A}}(\mathbf{r})$ are the donor and acceptor transition densities of the excitations of interest. For example, the transition density between states $|m\rangle$ and $|n\rangle$ of a system is defined as

$$\rho_{mn}(\mathbf{r}) = N \int ... \int \mathrm{d}\mathbf{r}_2 ... \mathrm{d}\mathbf{r}_N \Psi_m(\mathbf{r},\mathbf{r}_2, ..., \mathbf{r}_N) \Psi_n^{*}(\mathbf{r},\mathbf{r}_2, ..., \mathbf{r}_N) \tag{5}$$

where N is the number of electrons in the system. Since a transition occurs between two orthogonal states, integration of a transition density over the space $\mathbf{r}$ is zero. Therefore, a transition density contains an equal amount of positive and negative elements with a zero total charge. Under the dipole approximation, the Coulomb interaction (J_{coul}) reduces to the well-known Förster dipole−dipole coupling. The Coulomb coupling is generally regarded as the major component in EET coupling.

2.1. The Fragmented Density Approach. In GMH,[32] the dipole matrix elements are calculated in the eigenstate space. The charge-localized states are defined as the states that diagonalize the dipole moment operator. The ET coupling V_{RP} is then calculated as the off-diagonal matrix element of the Hamiltonian between two charge-localized states $|R\rangle$ and $|P\rangle$

$$V_{RP} = \frac{\mu_{mn} \Delta E_{mn}}{\sqrt{\Delta\mu_{mn}^2 + 4\mu_{mn}^2}} \tag{6}$$

where μ_{mn} is the transition dipole moment, ΔE_{mn} is the energy difference, and $\Delta\mu_{mn}$ is the permanent dipole moment difference, all of which are calculated between the initial and final (eigen) states ($|m\rangle$ and $|n\rangle$) of an ET reaction. Equation 6 was obtained by assuming that the difference in permanent dipole moments of the two states ($\Delta\mu_{mn}$) and the transition dipole moment (μ_{mn}) are parallel to each other.

In FCD,[33] the dipole moment operator in GMH is replaced by a difference charge operator. The molecule is partitioned into two fragments, one for the donor and the other for the acceptor. A 2 × 2 donor−acceptor charge difference matrix $\Delta\mathbf{q}$ is defined with its element Δq_{mn}

$$\Delta q_{mn} = \int_{r \in D} \rho_{mn}(r) \mathrm{d}r - \int_{r \in A} \rho_{mn}(r) \mathrm{d}r \tag{7}$$

where the diagonal elements Δq_{mm} are calculated from the diagonal one-particle density $\rho_{mm}(r)$, and the off-diagonal Δq_{mn} is the corresponding quantity from $\rho_{mn}(r)$, the transition density of states $|m\rangle$ and $|n\rangle$ (eq 5). Therefore, the matrix for charge difference can be calculated in eigenstate space

$$\Delta\mathbf{q} = \begin{pmatrix} \Delta q_{mm} & \Delta q_{mn} \\ \Delta q_{mn} & \Delta q_{nn} \end{pmatrix} \tag{8}$$

In general, $\Delta\mathbf{q}$ is not diagonal because transition density may exist between the two eigenstates. The charge-localized states are defined as the linear combination of the eigenstates that gives rise to a zero off-diagonal value for $\Delta\mathbf{q}$. It can be shown that this choice also maximizes the difference in the diagonal elements for $\Delta\mathbf{q}$. In this space, the off-diagonal Hamiltonian matrix element can be calculated as

$$V_{RP} = \frac{(E_m - E_n)|\Delta q_{mn}|}{\sqrt{(\Delta q_m - \Delta q_n)^2 + 4\Delta q_{mn}^2}} \tag{9}$$

For EET, a similar strategy can be employed. The excitations in eigenstates are usually slightly delocalized. The extent of such delocalization, plus the eigen energy difference, can, in principle, account for the EET coupling.[35] Instead of charges, we wish to define a quantity that counts the “excitation number”. An excitation can be viewed as the creation of an electron−hole pair. The number of such an electron and a hole can be calculated using the attachment and detachment densities, the positive and negative definite part, respectively, of the difference of the ground- and excited-state densities.[36]

In a single excitation model such as the configuration interaction singles (CIS), the wave function is written as

$$\Psi_n = \sum_{ia} a_{ia}^{(n)} \Phi_i^a \tag{10}$$

where Φ_i^a is a single substituted determinant constructed from the Hartree−Fock (HF) determinant Φ_0 by replacing an occupied orbital i with an unoccupied orbital a. With the CIS model, the difference density in the molecular orbital basis contains a negative occupied−occupied block and a positive virtual−virtual block, the absolute value of the former representing the detachment, or the "hole" population, and the latter the attachment, or the "electron" population, in an excitation.[36] They can be directly expanded in terms of the molecular orbitals $\{\psi_i(\mathbf{r})\}$

$$\rho_{\text{hole}}^{(n)}(\mathbf{r}) = \sum_{ij}^{\text{occ}}\sum_{a}^{\text{vir}} a_{ia}^{(n)}\, a_{ja}^{(n)*}\psi_i(\mathbf{r})\psi_j^*(\mathbf{r}) \tag{11}$$

and

$$\rho_{\text{elec}}^{(n)}(\mathbf{r}) = \sum_{ab}^{\text{vir}}\sum_{i}^{\text{occ}} a_{ia}^{(n)}\, a_{ib}^{(n)*}\psi_a(\mathbf{r})\psi_b^*(\mathbf{r}) \tag{12}$$

The electron and hole densities, eqs 11 and 12, can be used in the place of ρ_{nn} in eq 7 for the excitation population. We now need to define the corresponding off-diagonal density function for the off-diagonal matrix elements.

With the single-excitation theory, the state-to-state transition density as that in eq 5 can be expressed in terms of molecular orbital basis. It is similarly composed of an occupied−occupied and a virtual−virtual blocks. We can similarly define the hole transition density as the negative value of the occupied−occupied block, while the electron transition density is defined as the virtual−virtual block. Expanding the transition densities in these blocks yields the following

$$\rho_{\text{hole}}^{(mn)}(r) = \sum_{ij}^{\text{occ}}\sum_{a}^{\text{vir}} a_{ia}^{(m)}\, a_{ja}^{(n)*}\psi_i(r)\psi_j^*(r) \tag{13}$$

and

$$\rho_{\text{elec}}^{(mn)}(r) = \sum_{ab}^{\text{vir}}\sum_{i}^{\text{occ}} a_{ia}^{(m)}\, a_{ib}^{(n)*}\psi_a(r)\psi_b^*(r) \tag{14}$$

which are similar to those in eqs 11 and 12, except now, the off-diagonal densities of two excited states are involved. To find localized excitation, we wish to localize both the electron and hole distribution. It is therefore natural to use the sum of the electron and hole densities as a measure of excitation

$$\rho_{\text{ex}}^{(mn)}(r) \equiv \rho_{\text{hole}}^{(mn)}(r) + \rho_{\text{elec}}^{(mn)}(r) \tag{15}$$

For a pair of CIS excited states m and n, we can now define the quantity for EET corresponding to the difference charge Δq_{ij} in eq 7 as

$$\Delta x_{mn} = \int_{r\in D} \rho_{\text{ex}}^{(mn)}(r)\mathrm{d}r - \int_{r\in A} \rho_{\text{ex}}^{(mn)}(r)\mathrm{d}r \tag{16}$$

where the difference in the excitation population is now used to find the excitation localized states. The EET coupling can now be evaluated as

$$V_{RP} = \frac{(E_m - E_n)|\Delta x_{mn}|}{\sqrt{(\Delta x_m - \Delta x_n)^2 + 4\Delta x_{mn}^2}} \tag{17}$$

in which Δx_m and Δx_n are Δx_{mm} and Δx_{nn}, respectively. In the symmetrical situation (identical donor and acceptor), eq 17 becomes one-half of the energy gap ($E_m - E_n$), as used in the literature.[22,26,27]

The expression in eq 17 offers a new way to calculate the full EET coupling for a general class of systems irrespective of their symmetries. We propose to name this scheme the "fragment excitation difference" (FED) as a direct generalization from its preceding FCD scheme for ET couplings.[37] In FED, the diagonal excitation densities ρ_{ex}^{mm} are the sum of attachment and detachment densities, and the off-diagonal density matrix ρ_{ex}^{mn}, as shown in eqs 13 and 14, is simply the transition density between the states.

We first calculated ρ_{ex}^{mn} in the atomic orbital (AO) basis ($\mathbf{P}^{mn}$). To partition the densities to the donor and acceptor space, the Mulliken population analysis for the atomic fraction of the density was employed

$$\Delta x'_{mn} \equiv \sum_{\mu\in D}(\mathbf{P}^{mn}\mathbf{S})_{\mu\mu} - \sum_{\mu\in A}(\mathbf{P}^{mn}\mathbf{S})_{\mu\mu} \tag{18}$$

We note that the transition density is not symmetric in the Mulliken population analysis. We have followed the previous work on FCD[33] and employed a symmetrized definition for the excitation

$$\Delta x_{mn} \equiv \frac{1}{2}(\Delta x'_{mn} + \Delta x'_{nm}) \tag{19}$$

The FED scheme was implemented and calculated in a developmental version of Q-Chem.[38]

2.2. The Coulomb Coupling. To compare with the coupling values from the FED scheme, we have calculated the Coulomb coupling of the donor and acceptor for the separated intermolecular EET cases, as shown in eq 2.

In previous works,[23,24] the values of transition density in a three-dimensional grid were recorded, and a six-dimensional Riemann sum was performed for the Coulomb integration in eq 2. This approach requires a large amount of computer time, and the numerical quality is rather limited due to the steep cusp in the Coulomb kernel $1/r_{12}$. The Coulomb integration in eq 2 can be calculated using the analytical integration routine of atomic orbitals in quantum chemistry computation programs.[27,29] The transition density is the one-electron density operator for the ground state and an excited-state n

$$\rho(\mathbf{r}) = N\int \mathrm{d}r_2 \ldots \mathrm{d}r_N \Psi_n \Phi_0^* \tag{20}$$

With Ψ_n defined in eq 10, we can obtain

$$\rho(\mathbf{r}) = \sum_{ia} a_{ia}^{(n)}\psi_i^*(\mathbf{r})\psi_a(\mathbf{r}) = \sum_{ia}\sum_{\mu\nu} a_{ia}^{(n)}\, C_{\mu i}^* C_{\nu a}\phi_\mu^*(\mathbf{r})\phi_\nu(\mathbf{r}) \tag{21}$$

where ϕ_μ(r) is an atomic orbital and $C_{\mu i}$ is the coefficient in the linear combination for molecular orbitals. With excitations calculated separately in donor and acceptor fragments, we can rewrite the Coulomb integration of eq 2 as

$$J_{\text{coul}} = \sum_{iajb}\sum_{\mu\nu\lambda\sigma} a_{ia}^{\text{D}*}\, a_{jb}{}^{\text{A}} C_{\mu i} C_{\nu a}^*\, C_{\lambda j}^* C_{\sigma b}[\nu\mu|\lambda\sigma] \tag{22}$$

where the two-electron integral is defined as

$$[\nu\mu|\lambda\sigma] = \int \mathrm{d}\mathbf{r}_1 \int \mathrm{d}\mathbf{r}_2 \phi_\nu^*(\mathbf{r}_1)\phi_\mu(\mathbf{r}_1)\frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|}\phi_\lambda^*(\mathbf{r}_2)\phi_\sigma(\mathbf{r}_2) \tag{23}$$

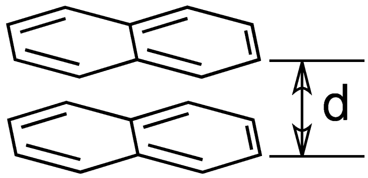

Figure 1. A pair of stacked naphthalenes employed to test for the EET coupling calculations.

The two-electron integrals in atomic basis functions are routinely calculated using analytical expressions in a quantum chemistry package. We took advantage of such integral codes and calculated the Coulomb coupling with eq 22, with much improved numerical quality and computational efficiency. For truncated models, we included the Coulomb coupling values to compare with the full EET values. All of the FED and Coulomb couplings reported in the present work were calculated using a developmental version of Q-Chem.[38]

3. Results and Discussion

We tested the new FED scheme in both through-space and through-bond EET between two naphthalenes that were studied extensively in the literature,[22,26,27,29,39] including a pair of stacked naphthalenes with varying distance and a series of rigidly linked naphthalene and anthracene dimers.

FED is a general scheme that does not depend on the level of the theory employed in an excited-state calculation. In principle, time-dependent density functional theory (TDDFT) can be used. However, for a pair of well-separated naphthalene molecules, we have observed unphysical results from TDDFT calculations where low-energy charge-transfer states were seen (data not shown). This is similar to a previous result, where TDDFT calculations were shown to yield qualitatively different results from those of ZINDO.[27] On the other hand, CIS results were similar to those reported with semiempirical models,[39] with unambiguous state assignments that are consistent with spectroscopy and computational literature. The only exception is that the first (^1L_b, or $^1B_{3u}$) and second (^1L_a, or $^1B_{2u}$) are reversed in their state ordering in a CIS Hamiltonian. While we are concerned about the excitonic couplings, the absolute position of the states does not cause a foreseeable problem in the computation. Therefore, all of the coupling values reported in the present work are based on the CIS Hamiltonian.

3.1. Dissecting the EET Couplings. We first tested the FED and the new Coulomb scheme with a pair of stacked naphthalene molecules as shown in Figure 1, which was previously studied.[26] Since for symmetric systems the FED coupling is trivially reduced to one-half of the energy gap, we used an asymmetric structure in this work for testing the new FED scheme, with one of the molecules optimized in its singlet ground state and the other optimized in the triplet state, both at the B3LYP/DZP level, unless stated otherwise.

The low-energy excited states of a naphthalene includes L_b (1^1B_{3u} or $^1B^-_{3u}$), L_a (1^1B_{2u} or $^1B^+_{2u}$), and B_b (2^1B_{3u} or $^1B^+_{3u}$). It is instructive to examine the various aspects of the coupling with one of the states first. In Figure 2, the EET coupling for the L_a state at a number of separations is shown. It is seen that at large separation, Coulomb coupling approaches the FED total coupling, and both decay as the expected d^{-3} dipole−dipole interaction.

The FED couplings reflect total coupling strengths, while the Coulomb coupling can be calculated precisely. As a result, the difference in these two coupling values reflects the short-range coupling, including the exchange coupling and the overlap term.

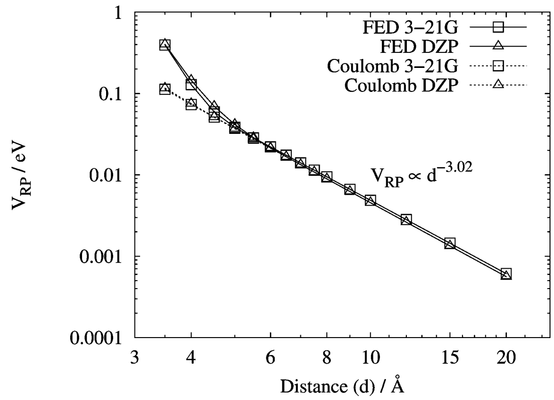

Figure 2. Full FED coupling and Coulomb coupling in two different basis sets, based on the L_a (1^1B_{2u}) state in CIS excitation. The fitted slope was obtained for distances at 8−20 Å for both FED and Coulomb coupling using DZP data.

As seen in Figure 2, the deviation of the FED and Coulomb couplings at a small separation is the short-range contribution. In Figure 3, the distance dependence of this difference is shown. Since we are subtracting two numbers with a finite numerical resolution, a numerical instability is seen when the value is small, particularly for results less than 10^{-4} eV. Nevertheless, as shown in Figure 3, the difference in couplings decays exponentially in a large basis set, and the exponent is very similar to that of a triplet−triplet (TT) energy transfer, an energy transfer coupling that does not contain the Coulomb coupling.[34] The difference in FED and Coulomb coupling decays at a rate of 2.60 Å^{-1} (derived from the data of 3.5−6 Å, with DZP+ basis set) or 2.85 Å^{-1} (DZP basis set). These values are very close to the exponent for TT coupling, which was fitted to be 2.60 Å^{-1} (CIS/DZP+) or 2.91 Å^{-1} (CIS/DZP). Moreover, for small basis sets, a Gaussian decay can be seen in the difference of FED and Coulomb couplings. The basis set dependence is also very similar to that of TT coupling. This result indicates that the difference of FED and Coulomb couplings behaves very much like an "exchange" component such as the TT energy-transfer coupling.

The FED and Coulomb couplings were calculated for the L_a, L_b, and B_b states, and their values were included in Figure 4. The FED couplings for the first two excited states, $L_a$and L_b, agree well with those reported previously,[26] presumably from CIS energy gaps in a symmetric structure. It is seen that for all three states, the Coulomb and FED couplings are essentially the same at large distances. At short distances, there exists a discrepancy arising mainly from exchange and overlap effects. A very important feature is that the Coulomb coupling does not always quickly approach the typical dipole−dipole d^{-3} distance dependence. It is seen that the Coulomb coupling for L_b decays faster than that of the other two states, while for B_b, the Coulomb coupling decays slower. This result indicates that it is insufficient to conclude the role of the short-range coupling simply from a deviation of dipole−dipole prediction.

In Table 1, we list the power of the Coulomb coupling with respect to the distance between two naphthalenes. It is seen that even at a distance of 10−20 Å, the Coulomb coupling may still significantly deviate from the d^{-3} decay. A multipole analysis indicates that the deviation is a result of multipole components of different signs (data not shown). An opposite sign reduces the major dipole−dipole coupling and leads to a smaller power. When the multipole component is of the same sign as the dipole−dipole term, the power is larger than 3.

3.2. EET Coupling in Rigidly Linked Molecules. The EET coupling in two rigidly linked naphthalene fragments has been

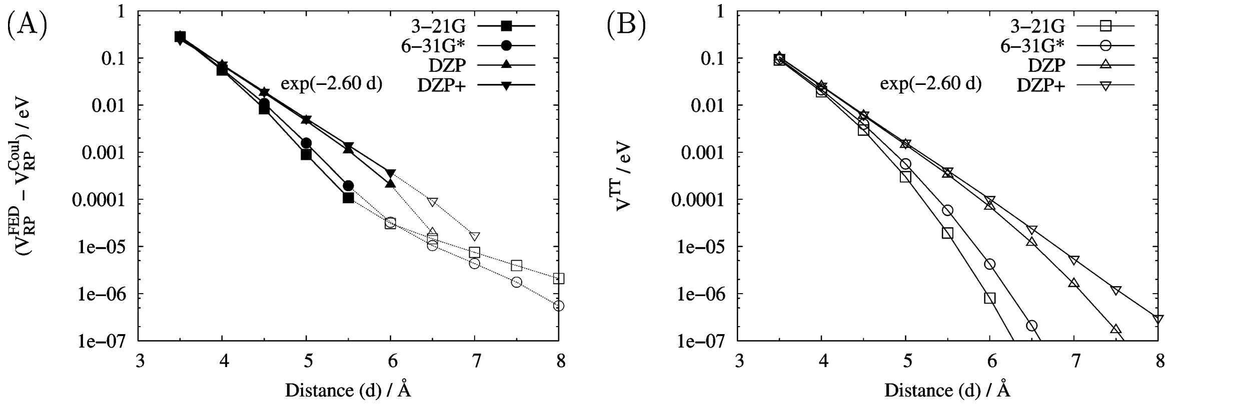

Figure 3. (A) Difference in the total FED coupling and Coulomb coupling values at a number of distances for the L_a (1^1B_{2u}) state between a pair of stacked naphthalenes. Thick lines and closed symbols were used for coupling values larger than 0.1 meV. For comparison, in panel (B) we included the TT energy-transfer coupling for the 1^3B_{2u} state of two naphthalenes. The fitted exponent 2.60 $Å^{-1}$ in both panels was obtained for the DZP+ basis set at distances of 3.5−6 Å.

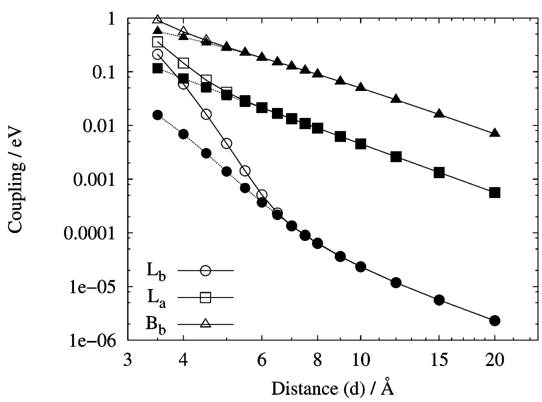

Figure 4. The FED coupling (open symbols with lines) and Coulomb coupling (filled symbols with dotted lines) values at a number of distances for L_b (circles), L_a (squares), and B_b (triangles) states.

TABLE 1: Fitted Powers of Coulomb Coupling with Respect to the Intermolecular Distances; Shown are Absolute Values of the Fitted Slopes on a log−log Scale

fitted range	L_b	L_a	B_b
3.5−6 Å	7.03	3.16	2.11
6.5−9 Å	5.42	3.02	2.54
10−20 Å	3.34	3.02	2.83

extensively studied.[22,27−29,39] The EET couplings as determined by the absorption spectral splitting in the B_b state in the polynorbornyl-linked naphthalene dimers, DN2, DN4, and DN6 (Figure 5), are far larger than that predicted with the dipole−dipole interaction. Therefore, a through-bond coupling component can be concluded.[1,22,27−29,39] Since there exists a symmetry in the DN series, the two excitonic states are either symmetric or antisymmetric combinations of the locally excited states. The EET coupling can be calculated as one-half of the energy gap. With both through-space and through-bond models, Clayton et al.[22] found a large increase in EET coupling due to the polynorbornyl bridge fragment. With a semiempirical INDO/S Hamiltonian, Tretiak et al.[39] calculated the collective electronic oscillators (CEO) of the DN molecules. The absorption spectra can be quite accurately simulated, and a strong coherence in transitions was observed in DN2. Solvent effects were included in a more recent work,[27] and the through-bond component was dissected using a generalized Harcourt model.[29] The through-bond nature of the EET coupling in these molecules has been well studied, and the role of short-range coupling has been discussed in light of multipole interaction.[26] By providing

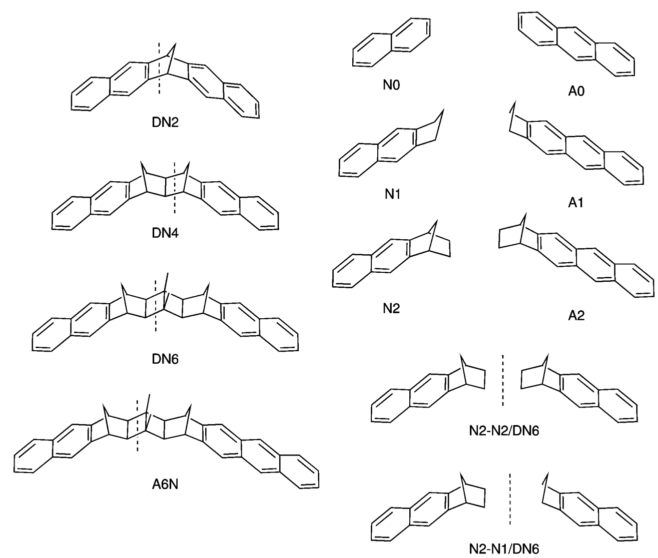

Figure 5. The polynorbornyl-linked dimers studied in the present work (DN2, DN4, DN6, and A6N). Truncated models are sketched to the right. Monomeric fragments are included in the top right. For DN2, only the N0−N0 model was considered. For DN4, we have considered N1−N1, N1−N0, and N0−N0 models by holding the naphthalene fragments fixed and sequentially removing bridge fragments in between. N2−N2, N2−N1, N1−N1, and N0−N0 models were considered for the DN6, and similarly, N2-A2, N2-A1, N1-A2, N1-A1, and N0-A0 models were considered for A6N. As examples, the structures of N2−N2/DN6 (the truncated model composed of two N2's based on DN6) and N2−N1/DN6 are shown. The vertical dashed lines indicate how we separated the molecules into the donor and acceptor fragments in FED calculations.

Coulomb coupling values in good numerical quality, we can obtain further insight into the role of short-range coupling in EET.

3.2.1. Through-Bond Effects. To probe the through-bond effects, we have included several truncated models, as shown in Figure 5. The FED scheme is applicable to all of the models, irrespective of their molecular symmetry. The Coulomb coupling can be obtained for systems composed of separated fragments because the donor and acceptor excitations have to be individually calculated. For the rigidly linked molecules studied, the donor fragments were optimized on the CIS/6-31G* S_1 state. The bridge and acceptor fragments were optimized at their ground state, namely, the B3LYP/6-31G* level. To mimic the structure of the molecules when EET takes place, we connected the excited donor structure with other parts optimized in their

TABLE 2: FED and Coulomb (Coul.) Coupling (in meVs) of the Molecules Studied

		L_b (S_1, 1^1B_{3u})		L_a (S_2, 1^1B_{2u})		B_b (S_3, 2^1B_{3u})	
molecules	models	FED[a]	Coul.[b]	FED	Coul.	FED	Coul.
DN2	full	60.1	–	88.3	–	521 (397)[c]	–
	N0–N0	93.8[d]	7.21	115[d]	37.7	595[d]	403
DN4	full	14.0	–	20.9	–	269 (221)[c]	–
	N1–N1	9.75	2.99	13.8	8.13	218	191
	N1–N0	1.54	1.22	7.36	6.20	188	174
	N0–N0	0.703	0.604	5.06	4.92	151	149
DN6	full	2.28	–	4.47	–	143 (99)[c]	–
	N2–N2	2.41[d]	1.35	4.02	2.56	132	121
	N2–N1	1.64	1.18	2.56	2.17	113	107
	N1–N1	1.10	0.995	1.75	1.69	91.2	89.1
	N0–N0	0.103	0.105	1.83	1.83	65.4	65.4
A6N	full	1.95	–	4.09	–		
	N2-A2	2.11[d]	1.27	3.74	2.99		
	N2-A1	1.41		2.81			
	N1-A2	1.38		2.84			
	N1-A1	1.03		2.37			
	N0-A0	0.096	0.096	2.43	2.43		

[a] Coupling calculated via the new FED scheme based on CIS/6-31G* excitations. [b] Coulomb coupling of the donor–acceptor excitations for the truncated models indicated. [c] In the parentheses are the experimental values as a half spectral splitting as reported in ref 28. [d] Exceedingly large values due to closely spaced truncated models (see text).

ground state. Further optimizing the full molecular structure in their donor excited state does not change the result significantly, as the Condon approximation implies.[40] The positions of H atoms added to cap broken bonds were optimized in the ground state at the B3LYP/6-31G* level for N1–N1 of DN4, N2–N2 of DN6, and N2-A2 of A6N.[41] The structures of the molecules studied are included in the Supporting Information accompanying this work. In Table 2 and Figure 6, we included the FED and Coulomb coupling values of the full DN series and their corresponding truncated models.

For all of the three states calculated in linked naphthalene dimers, as shown in Figure 6, the coupling values increase with more bridge fragments restored in the truncated models, and thus, a through-bond EET coupling can be concluded. In all of the cases studied, the through-bond EET coupling contains a significant Coulomb effect, as can be seen by an increase of Coulomb coupling with larger bridge fragments in the model, for naphthalene dimers as well as for A6N. This result is consistent with previous reports.[27,29]

The difference between the FED and Coulomb couplings gives us an idea of the short-range coupling magnitude. Such short-range couplings are typically smaller than the Coulomb couplings in most cases listed in Table 2 or Figure 6, except for very closely spaced truncated models, N0–N0 of DN2, N1–N1 of DN4, N2–N2 of DN6, and N2-A2 of A6N. In these closely spaced models, some of the FED couplings are larger than those of their corresponding full models, as indicated by asterisks in Figure 6. The short-range couplings are also increased significantly in these closely spaced models, as inferred by the large gap between the FED and Coulomb coupling values. Therefore, we believe that the larger FED coupling values in the closely spaced models are from orbital overlaps of the two fragments.[41] Therefore, the FED coupling values in these very closely spaced models may not represent a reasonable estimate to infer the role of the truncated bridge in EET coupling, but their corresponding Coulomb coupling values may still be meaningful since they depend weakly on the overlap of wave functions. We chose to use asymmetric models such as N1–N0 for DN4 and N2–N1 for DN6 (Figure 6) as alternative closely spaced truncated models.

In Table 3, we listed the estimated percentage of through-bond contribution and ranges for possible short-range couplings. The through-bond component was estimated from the difference of FED coupling in the full and the minimal (N0–N0) truncated models. For the short-range coupling, it would be best to obtain such a quantity for the full molecules, but the Coulomb coupling value for a fully linked molecule is not available yet. We used the largest truncated model values as a substitution. This led to a minimum estimate for the Coulomb coupling and therefore the largest possible short-range coupling. For an estimate on the lower bound of the short-range coupling, we simply took the difference of the FED and Coulomb couplings in the largest truncated models that are free from unphysically large FED coupling. With the truncation, we believe that the short-range couplings are reduced, and therefore they can serve as the lower bound for the unknown value in the full molecule.

From Table 3, it is seen that the through-bond couplings are very significant parts of the EET coupling. In all cases, they constitute more than 40% of the overall EET coupling, and for the L_b state, it reaches 95% in all three cases studied. The L_b state has a weak transition strength. In the alternacy symmetry,[42] it is a forbidden state (the L_b state is a "minus" state, ${}^1B_{3u}^-$, while the ground state is also a minus state, ${}^1A_g^-$).[42,43] The weak transition strength of an isolated naphthalene arises because the Fock matrix does not follow the alternacy symmetry exactly—there exist interactions beyond nearest-neighboring atoms. When the naphthalene is attached to a bridge fragment, the perturbation may enhance the weak transition by further symmetry breaking. The transition dipole of this L_b state in a monomer naphthalene is 0.194 Debye, and it is increased to 0.785 or 0.842 Debye in bridge-attached N1 or N2. With the dipole strength increasing by a factor of 4, the dipole–dipole coupling is increased by a factor of 16 for naphthalene dimers, which is close to the factor of 20 observed in the EET coupling. Thereby, the large through-bond EET coupling in this case mainly arises from a large increase in the transition dipole.

The EET couplings of the L_a and B_b states are increased by a factor of 2 or 3 with the polynorbornyl linking bridge. These two states are allowed states in the alternation symmetry (The L_a state is a "plus" state, ${}^1B_{2u}^+$, while the B_b state is also a plus state, ${}^1B_{3u}^+$.). Their transition dipoles are only slightly modified by the presence of the bridge fragments (data not shown). Nevertheless, the overall Coulomb couplings are increased, possibly due to higher multipole effects.

3.2.2. The Short-Range Couplings. In Table 3, estimates on the upper and lower bounds of the short-range coupling are listed. It is seen that the contribution from the short-range coupling can be very large, especially for the L_a and L_b states.

In Figure 6A, it is seen that for the bright B_b state, most of the through-bond component is Coulomb in nature because there is a very close agreement between Coulomb and FED couplings. In DN4 and DN6, the EET coupling values steadily increase as the size of bridge fragments increase. The Coulomb couplings of N2–N2/DN6 and N1–N1/DN4 are roughly in the middle of the FED coupling of the full model and that of smaller models, N2–N1/DN6 or N1–N0/DN4. If the Coulomb coupling of the full DN4 or DN6 could have been calculated, it would probably have been somewhat higher than that of the N2–N2/DN6 or N1–N1/DN4 models, similar to the FED coupling values (of the full molecules), as roughly projected by the overall trend of the Coulomb data. This result implies that Coulomb coupling dominates the overall coupling in this bright B_b state, and the short-range coupling is a minor contribution. This result

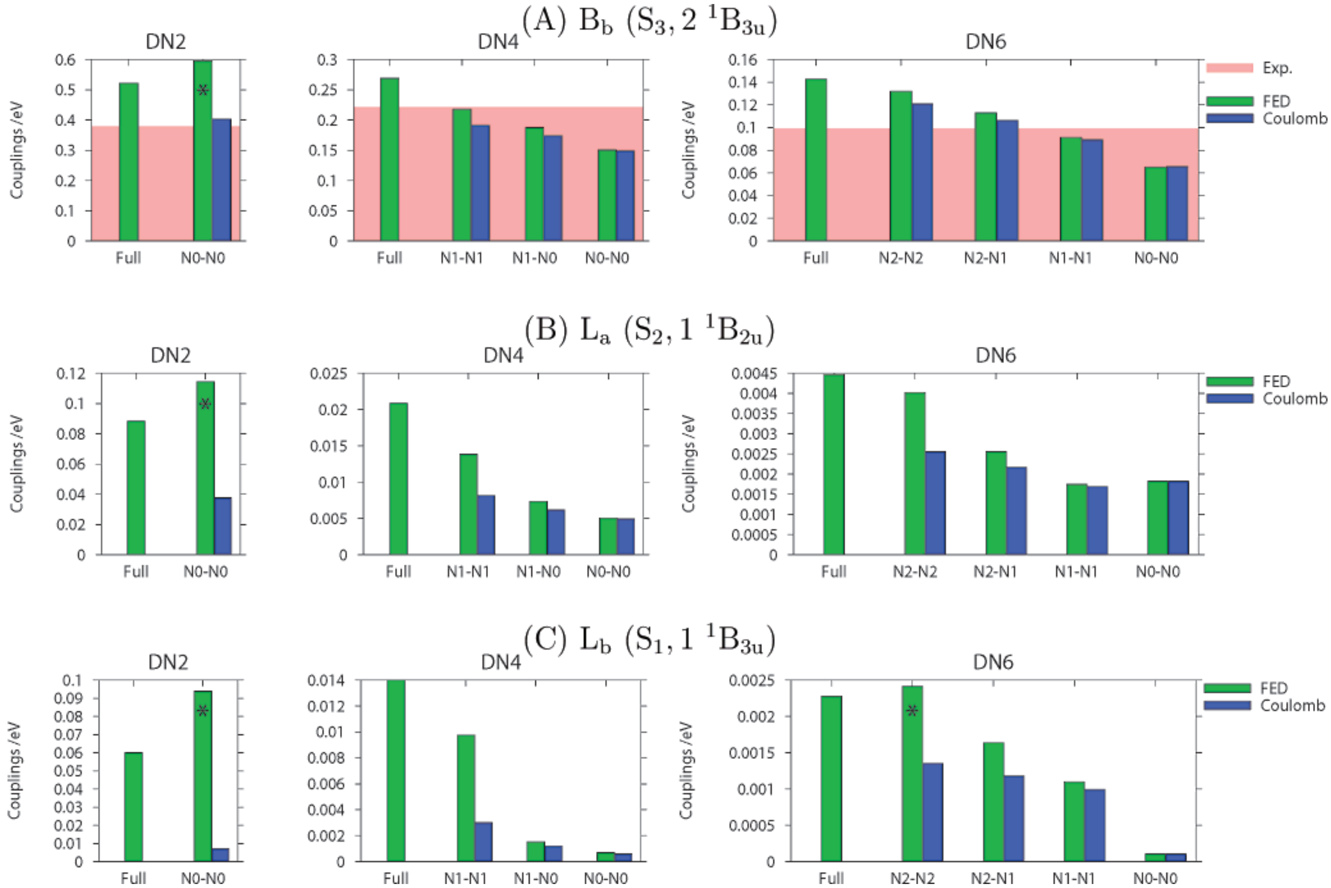

Figure 6. EET couplings of the DN series, calculated with FED and Coulomb schemes for the lowest three singlet excited states of naphthalenes, and in panel (A), experimental values are included as the shaded background. Asterisks denote FED results that may suffer from an artifact of a crowded truncated model (see text).

TABLE 3: Percentage of the Through-Bond (TB) Contribution and the Maximum/Minimum Possible Short-Range (SR) Coupling in the Total EET Coupling

	TB %[a]			max. SR %[b]			min. SR %[c]		
	L_b	L_a	B_b	L_b	L_a	B_b	L_b	L_a	B_b
DN2	–	–	–	88	57	23			
DN4	95	76	44	79	61	29	21	16	7.3
DN6	95	59	54	41	43	15	28	15	5.5
A6N	95	41	–	35	37	–			

[a] Estimated as $(V^{\mathrm{FED}}_{\mathrm{Full}} - V^{\mathrm{FED}}_{\mathrm{N0-N0}})/V^{\mathrm{FED}}_{\mathrm{Full}}$, the difference in the FED coupling of the full molecule and the N0−N0 truncated model, relative to the FED coupling of the full molecule. For DN2, the N0−N0 truncated model is too crowded, and therefore, the FED coupling was regarded as an unreliable result. [b] Maximum possible short-range contribution estimated as $(V^{\mathrm{FED}}_{\mathrm{Full}} - V^{\mathrm{Coul}}_{\mathrm{Ni-Ni}})/V^{\mathrm{FED}}_{\mathrm{Full}}$, the difference of the FED coupling of the full molecule and the Coulomb coupling of the largest truncated models ($i = 2$ for DN6, $i = 1$ for DN4, or $i = 0$ for DN2). Reported are percentages relative to the FED coupling values in the full molecules. [c] Minimum possible short-range contribution estimated as $(V^{\mathrm{FED}}_{\mathrm{Ni-N[i-1]}} - V^{\mathrm{Coul}}_{\mathrm{Ni-N[i-1]}})/V^{\mathrm{FED}}_{\mathrm{Ni-N[i-1]}}$, the difference of the FED and Coulomb coupling in N1−N0/DN4 or N2−N1/DN6 models, the largest truncated models without an unphysically large FED coupling.

can also be seen directly in the estimated range, 5.5−29%, for short-range coupling (Table 3).

For L_b and L_a states in the truncated models, most of the FED couplings are also quite similar to the Coulomb coupling, except for the closely packed cases, as discussed above. For DN4 and DN6, the trend in the Coulomb coupling seems to end at a lower value than that of the full molecules. For the N2−N1/DN6, there exists a small but visible difference in the FED and Coulomb coupling for both L_a and L_b states. These results imply that the EET coupling in these states contains a visible, nontrivial short-range component, which is also seen in the results listed in Table 3. From our rough estimation, the short-range coupling may contribute 21−88% in the L_b state and 15−61% in the L_a state.

3.2.3. Additional Remarks. Compared to experimental results on the B_b state splitting, our calculated FED couplings are about 20−40% larger. The Hamiltonian that we based our results on is CIS, which is known to overestimate transition dipoles. When the transition densities are scaled down, the Coulomb couplings calculated give reasonable estimates for the light-harvesting rates in LH2.[23] Therefore, the CIS Hamiltonian for the EET coupling is generally acceptable, but the Coulomb interaction can be overestimated. Another possible cause of this discrepancy would be the solvent effects. In a previous work, it was found that the solvent reduced the EET coupling by similar percentages in the linked naphthalenes.[27]

For molecules of similar sizes as those studied in the present work, TDDFT would be another important tool. Our FED scheme can be generalized to TDDFT, and it would be interesting to test for the exchange-correlation effect on the EET coupling. However, as mentioned above, for naphthalene dimers, we found that TDDFT yields low-lying charge-transfer states that involve exchange of electrons of the two naphthalenes, making it difficult to assign the low-lying states confidently.

Another note we make is for the different definitions of "through-bond" contribution in EET coupling. In the Harcourt model,[44] the locally excited components are separated from "ionic", charge-transfer excitations, and, thus it is natural to define the covalent contribution as that from locally excited components, while others are defined as a "through-bond" effect. Such a definition was used in many previous works.[26,29,45] The covalent contribution versus the through-bond effects of the same set of (poly)norbornane-spaced molecules were studied under the influence of solvent polarizability.[29] However, as can be seen in Figure 3 of ref 29, the molecular orbitals after the linear transformation may still be slightly delocalized to the bridge fragments. Such a bridge-mediated effect may contribute to the covalent contribution in the Harcourt model, and it is not regarded as a through-bond effect. In a truncated model, however, all of the effects of such delocalized orbitals are included in the through-bond effects. Therefore, the definition of the through-bond effect with the Harcourt model is different from the definition employed in the present work. We thought it would be the easiest to use truncated models to access the through-bond effect. While it may become hard to correlate terms of different origins because now the molecular orbitals are different in different truncated models, we found that it is intuitively more clear to adapt this through-bond model.

The FED coupling and the Harcourt model have also yielded different results for the closely spaced models. In ref 29, the couplings obtained from the Harcourt model in the full molecules were larger than those in the closely spaced models, while in our case, the FED total couplings in spaced models are sometimes larger than those of full molecules. The Harcourt model is probably less prone to the problem of large short-range interaction because it includes only a small number of valence excitations (covalent and ionic). We speculate that the orbital overlap would distort the active molecular orbitals slightly, leading to possibly small populations in many higher excitation configurations in the excited states that are not accounted in the Harcourt model.

By comparing the coupling in full molecules and their corresponding truncated models, we can obtain the total through-bond effect. With the Coulomb coupling calculated to a numerical quality similar to that of the full EET coupling, we can infer the importance of Coulomb coupling and the short-range coupling in this through-bond component. In the cases we studied, Coulomb coupling plays an important role in all cases, while the contribution of the short-range coupling varies.

4. Conclusion

The newly developed FED scheme can be used to calculate EET coupling in a general class of molecules. With it, and an improved Coulomb coupling scheme, we characterized the short-range coupling component in EET with a number of naphthalene-containing systems studied before. We show that the short-range component is very similar to the TT energy-transfer coupling, a spin-exchange case of energy transfer that does not contain Coulomb coupling. For rigidly linked donor−acceptor systems, a significant through-bond contribution can be seen, and the role of short-range coupling was investigated.

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Supporting Information Available: Geometries of studied molecules. This material is available free of charge via the Internet at http://pubs.acs.org.

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